

Thiruwaran Kalvin

University Of Liverpool • [linkedin.com/in/t-kalvin](https://www.linkedin.com/in/t-kalvin) • github.com/t-kalv • Website

PROFESSIONAL SUMMARY

Current second year Computer Science with Artificial Intelligence student reading at the University of Liverpool with a strong interest in AI Research, Robotics, Reinforcement Learning and Intelligent Systems. Working towards a long-term ambition of pursuing a career as an Artificial Intelligence Researcher and Robotics Engineer, contributing to world-leading advancements in autonomous systems and applied robotics. Experienced in hands-on experimentation through open-source projects and research initiatives, including developing adversarial attack and defence algorithms from scratch, applied computer vision systems and real-time software pipelines. Dedicated to professional development, teamwork and engaging meaningfully with diverse engineering teams and researchers in fast-paced, collaborative research environments.

EDUCATION

University Of Liverpool

Liverpool, United Kingdom

BSc Hons, Computer Science with Artificial Intelligence

September 2024 - Present

Classification: First-Class (Predicted Grade)

Relevant Coursework: Advanced Artificial Intelligence, Introduction to Data Science, Deep Learning, Introduction to Artificial Intelligence, Object-orientated Programming, Software Engineering, Data Structures and Algorithms, Complexity of Algorithms, Principles of C & Memory Management, The C++ Programming Language, Computer Systems, Scripting Languages, Database Development, Programming Language Paradigms, Group Software Project, Analytic Techniques for Computer Science, Computer-based Trading in Financial Markets

WORK EXPERIENCE

LASER (Liverpool Association For Space Engineering And Research)

Liverpool, United Kingdom

Rocket Team – ML Software Engineer And Project Lead (Part-Time)

(October 2024 – Present)

- Developed the **VEGA multispectral imaging rideshare payload** using a Raspberry Pi Zero 2W and a modified Raspberry Pi NoIR Camera Module 3, implementing a **real-time edge-computing computer vision/machine learning algorithm** that calculates and **classifies vegetation health live**.
- Engineered a **custom real-time Ground Station UI (Python, PyQt, PySide6, PyQtGraph, Matplotlib)**, enabling **live 3D rocket attitude visualisation** via quaternions and **dynamic 2D plotting of critical telemetry** data received from the **LIFTS v2 onboard avionics system**.
- Programmed **high-speed C++ firmware** for an **Adafruit Feather RP2040** and **LoRa (RFM95) transceiver**, optimising telemetry reception with a **500kHz bandwidth** and implementing **custom buffer-flushing** to prevent data ghosting.
- Designed **custom PLOTS ground station** electronics designed in **KiCad** featuring **Adafruit Feather RP2040**, **Raspberry Pi 5** and **Waveshare Touchscreen display**.
- Implemented **Python** and **Adafruit's documentation** to interface with multiple sensors (BMP390, MPU6050, H3LIS331) **enabling precise data collection** for obtaining rocket telemetry.
- Influenced the **design and schematics** of the LIFTS flight computer, **integrating 5+ electrical components and sensors, improving overall optimal system performance by 30%**.
- Co-Lead a multidisciplinary team of 10+ engineers** to **design, build and innovate a state-of-the-art vehicle** capable of achieving its intended altitude, recovery outcomes and rideshare experiment, fostering a culture of **innovation** through **shared ideas** and **problem solving**.
- Conducted thorough **research through documentation** and **comparative analysis of 10+ critical components**, including **sensors and microcontrollers**, to **inform design decisions and enhance system reliability**.

Retail Guys Ltd

Manchester, United Kingdom

IT Consultancy And Computer Technician (Part-Time)

(September 2021 – Present)

- Executed crucial roles **including software/hardware installation for over 100+ systems**, video tutorial creation, **database entry for 500+ records** and consultancy.
- Recorded **20+ app technical documentation** and **video tutorials**, communicating with clients/users in a straightforward manner, regardless of technical knowledge.
- Analysed client interactions to identify potential system manipulations, **preventing 3+ unauthorized access to applications outside the main software**.
- Actively participated in **product testing, feedback cycles** and **workflow optimisation** within client organisations.

PROJECTS

VEGA (Vegetation Evaluation from Ground to Air) | Computer Vision Research Project/Research Paper.

September 2025 – Present

- Developing a **real-time Python computer vision payload** utilising **OpenCV** and **NumPy** to **capture and process aerial imagery** via a modified Raspberry Pi Camera Module 3 NoIR.
- Engineering a **custom visible-spectrum NDVI algorithm**, applying **dynamic auto-calibration** and **temporal smoothing** to **evaluate vegetation health** under varying environmental conditions.
- Implementing a **live HUD overlay** featuring **topographical contour of mapping healthy and unhealthy zones** in real-time, synchronised with an **automated logger** to record system telemetry and sensor data to CSV and MP4 formats.

Simple MNIST Digit Classifier Neural Network | Deep Learning Research Project/Research Paper.

July 2025 – September 2025

- Researched and designed a **Python Neural Network** from scratch to **recognise handwritten numerical digits** from the **MNIST dataset**, utilising **7960 trainable parameters** with Forward/Backpropagation, ReLU, SoftMax activation functions.

Interactive Black-Scholes Option Pricing Model | Website | Source Code.

August 2025

- Applied a Black-Scholes **Option Pricing model** for modelling **European options** using **Python Backend** and **Streamlit Frontend**, **Yahoo Finance API** fetching **near real-time quotes from 10,000+ global tickers** and **SQLite Backend** that **calculates option prices, estimates implied volatility and visualises data**.

Formula 1 Driver Stats | Website | Source Code.

August 2025

- Applied **F1 Driver Dataset** to produce a **full-stack Java Web App** with a **Spring-Boot, PostgreSQL Backend** containing **data from 25+ drivers with associated information** and **HTML/JS Frontend** providing **information** about the 2024 season **via a search interface** using **REST APIs**

TASP-TALIA | Python CLI/Voice App | Source Code.

July 2022 - Present

- Developed **TASP (Totally Auto Stock Pinger)** to notify user when tracked stocks reach specified buy/sell conditions, **pulling live data from Yahoo Finance**.
- Implemented **TALIA (Totally Artificial Language Intelligence Assistant)**, a **Python CLI/Voice assistant** that can **process typed or spoken 50+ commands** and assisting users with **everyday tasks**, leveraging **neuralintents.GenricAssistant** with **pre-trained intents model to map natural-language commands to corresponding functions, integrating APIs/services for live, real-time information**.

AI Python Draughts System | Python App | Source Code.

2023

- Applied an **artificial intelligence draughts (checkers) engine** for **Human vs Human/AI** using **Python** an **Pygame interface** for board visualisation on a **(10 x 10) board** handling gameplay with **40 pieces** and user interaction.
- Designed the **decision-making logic using the minimax algorithm** with **heuristic evaluation** functions demonstrating **applied game-AI algorithms**.

TECHNICAL SKILLS

- **Languages:** Python, Java, SQL, C#, C, C++ HTML, CSS, PHP, JavaScript, MATLAB, Haskell, Markdown, LaTeX
- **Developer Tools:** GitHub, Git, Linux, KiCad, VS Code, Visual Studio, IntelliJ, CLion, WebStorm, Jupyter Notebook, Docker
- **Libraries/Frameworks:** Streamlit, SQLite, PyQt, MySQL, MariaDB, OpenCV, PyTorch, NumPy, Pandas, Matplotlib, Spring Boot, Swing, LLMs

EXTRACURRICULAR/INTERESTS

- **Kaggle Certifications (Intro to Machine Learning, Deep Learning, AI Ethics and SQL)** – Gained practical experience in data analysis, model optimisation and responsible AI practices through hands-on real-world problems and real-world datasets.
- **Artificial Intelligence & Machine Learning Research** – Independent Research Projects such as handwritten numerical digit classifier using Python and Mathematics, enhancing research experience.
- **Blogging on latest advancements in AI & Computer Science** – Author of a personal blog explaining Research Papers, Projects and Concepts in the field of Computer Science, strengthening communication and subject expertise.
- **Competitive Programming** – Competed in online coding challenges and contests for enjoyment, improving algorithmic thinking, time-constrained problem-solving and programming efficiency.

REFERENCES

References Available upon Request.